Table S1: 26 Methods details.

No.	Methods	Classification	Implementation Principle	Implementation language	Available URL	Year of publication	Reference
1	PCA	Linear and Probabilistic Factor Models	PCA achieves dimensionality reduction by projecting high-dimensional data onto low-dimensional principal components with maximum variance through linear transformation	R	stats 4.3.3	1987	(Greenacre et al., n.d.)
2	GLM PCA	Linear and Probabilistic Factor Models	GLMPCA models raw counts using Poisson or negative binomial likelihoods and employs bias	R	https://github.com/xzhoulab/DRComparison/tree/master/data	2019	(Townes et al., 2019)

			residuals for rapid approximate computation				
3	pCMF	Linear and Probabilistic Factor Models	pCMF achieves dimensionality reduction by probabilistically decomposing data through a sparse Gamma-Poisson factor model	R	https://github.com/xzhoulab/DRComparison/tree/master/data	2019	(Durif et al., n.d.)
4	ZIFA	Linear and Probabilistic Factor Models	ZIFA maps high-dimensional data to a low-dimensional space by introducing factor analysis with a zero-inflation model	python	https://github.com/xzhoulab/DRComparison/tree/master/data	2016	("ZIFA: Dimensionality reduction for zero-inflated single-cell gene expression analysis," n.d.)

5	scGBM	Linear and Probabilistic Factor Models	scGBM directly models data through a Poisson bilinear model, achieving dimensionality reduction by introducing a low-dimensional factor structure in the log-mean representation	R	https://github.com/phillipnicol/scGBM	2023	(Nicol and Miller, 2023)
6	SSNMDI	Linear and Probabilistic Factor Models	SSNMDI projects high-dimensional data onto a low-dimensional space through non-negative matrix factorization, integrating data interpolation and label information within the framework to optimize	MATLAB	https://github.com/yushanqiu/SSNMDI	2023	(Qiu et al., 2023)

			dimensionality reduction				
7	tGPLVM	Linear and Probabilistic Factor Models	tGPLVM maps high-dimensional data to low dimensions via Gaussian processes and models noise using a heavy-tailed t-distribution	python	https://github.com/architverma1/tGPLVM	2020	(Verma and Engelhardt, 2020)
8	VAE	Deep Autoencoders and Generative Models	VAE maps high-dimensional data to a low-dimensional latent distribution via an encoder, then reconstructs it using a decoder to maximize the ELBO for	python	https://github.com/greenelab/CZI-Latent-Assessment/tree/master/single_cell_analysis	2019	(Hu and Greene, 2019)

			dimensionality reduction				
9	scvis	Deep Autoencoders and Generative Models	scvis combines variational autoencoders with t-SNE regularization to learn a probabilistic mapping function for dimensionality reduction	python	https://bitbucket.org/jerry00/scvis-dev	2018	(Ding et al., 2018)
10	VASC	Deep Autoencoders and Generative Models	VASC employs a deep variational autoencoder for dimensionality reduction and utilizes a ZI layer to handle dropout events	python	https://github.com/wang-research/VASC	2018	(Wang and Gu, 2018)

11	SAUCIE	Deep Autoencoders and Generative Models	SAUCIE employs a deep autoencoder with multiple regularization terms to simultaneously accomplish tasks such as dimensionality reduction, denoising, and clustering	python	https://github.com/KrishnaswamyLab/SAUCIE	2019	(Amodio et al., 2019)
12	scScope	Deep Autoencoders and Generative Models	scScope employs a deep recurrent autoencoder to iteratively repair missing values in scRNA-seq data while simultaneously performing batch normalization and feature learning	python	https://github.com/xzhoulab/DRComparison/tree/master/data	2019	(Deng et al., 2019)

13	SCD RHA	Deep Autoencoders and Generative Models	SCDRHA first applies DCA for denoising, then employs a graph attention autoencoder for dimensionality reduction	pytho n	https://github.com/WHY-17/SCDRH A	2021	(Zhao et al., 2021)
14	scGA E	Deep Autoencoders and Generative Models	scGAE achieves dimensionality reduction by constructing cell maps and using graph autoencoders	pytho n	https://github.com/ZixiangLuo1161/s cGAE	2021	(Luo et al., 2021)
15	DRE AM	Deep Autoencoders and Generative Models	DREAM employs a combination of variational autoencoders and mixture of Gaussians for dimensionality reduction of scRNA-seq data, utilizing a ZI layer to handle	pytho n	https://github.com/Crystal-JJ/DREA M	2023	(Jiang et al., 2023)

			dropout events				
16	DRA	Deep Autoencoders and Generative Models	DRA employs adversarial variational autoencoders for dimensionality reduction	python	https://github.com/eugenelin1/DRA	2020	(Lin et al., 2020)
17	UMAP	Graph-Based and Diffusion Geometry Models	UMAP achieves nonlinear dimensionality reduction by constructing a proximity graph in high-dimensional space and optimizing low-dimensional embeddings to preserve its topological structure.	python	https://github.com/lmcinnes/umap	2018	(Becht et al., 2019)

18	SIMLR	Graph-Based and Diffusion Geometry Models	SIMLR adaptively learns the optimal similarity matrix between cells through multi-core learning, utilizing this matrix for dimensionality reduction.	R	https://github.com/BatzoglouLabSU/SIMLR	2017	(B. Wang et al., n.d.)
19	PHATE	Graph-Based and Diffusion Geometry Models	PHATE performs dimensionality reduction through multidimensional scaling analysis while preserving topological information by constructing the diffusion geometry of the data.	python	https://github.com/KrishnaswamyLab/PHATE	2017	(Moon et al., 2019)

20	EDGE	Graph-Based and Diffusion Geometry Models	EDGE constructs an inter-cell similarity matrix by integrating numerous weak learners and performs dimensionality reduction based on this matrix.	R	https://github.com/shawnstat/EDGE	2020	(Sun et al., 2020)
21	SPDR	Graph-Based and Diffusion Geometry Models	SPDR achieves dimension reduction that preserves data structure while correcting batch effects by cross-batch sampling triplets and optimizing the loss function.	python	https://github.com/eleozzr/SPDR	2023	(Xu and Li, 2023)
22	IVIS	Metric Learning and Structure-Aware Embedding	Ivis employs a concatenated neural network and triplet loss to map	python	https://github.com/beringresearch/ivis	2019	(Szubert et al., 2019)

23	PaC MAP	Metric Learning and Structure-Aware Embedding	high-dimensional data to low dimensions while supporting rapid embedding of new data points. PaCMAP preserves the topological structure of data while performing dimensionality reduction by constructing three types of point pairs: nearest neighbors, intermediate neighbors, and distant neighbors.	pytho n	https://github.com/YingfanWang/Pa CMAP	2021	(Y. Wang et al., n.d.)
24	t-SN E	Metric Learning and Structure-Aware Embedding	t-SNE transforms the similarity between high-dimensional points into probability	R	Rtsne 0.17	2008	(Hinton and Roweis, n.d.)

			distributions and matches these distributions in a low-dimensional space to achieve dimensionality reduction while preserving structural information.				
25	TriMap	Metric Learning and Structure-Aware Embedding	TriMap learns low-dimensional embeddings by sampling triplet constraints from high-dimensional data and optimizing the loss function.	python	https://github.com/eamid/trimap?tab=readme-ov-file	2019	(Amid and Warmuth, 2022)
26	SQuaD-MDS	Metric Learning and Structure-Aware Embedding	SQuadMDS randomly groups data into quadruplets and utilizes gradient descent to optimize relative	python	https://github.com/PierreLambert3/SQuaD-MDS-and-FItSNE-hybrid	2022	(Lambert et al., 2022)

distance error,
achieving
multidimensional
scaling with
linear complexity.

Table S2: Real Datasets details.

Dataset Name/ID	Source Repository (Link/ID)	Species	Tissue/Cell Type	Condition (Health, Disease, Treatment)	Cell Number	Sequencing Tech	date
yan	GSE36552	Human	Embryonic stem cells	Developmental	90	Tang	2013
Baron	GSE84133	Human、Mouse	Pancreas	Healthy	8569、1886	inDrop	2016
Xin	GSE81608	Human	Islet Cells	Disease	1600	SMARTer	2016
Biase	GSE57249	Mouse	Embryonic cells	Developmental	56	SMARTer	2014
Zeisel	GSE60361	Mouse	Brain	Healthy	3005	STRT-Seq	2015
Goolam	E-MTAB-3321	Mouse	Embryonic cells	Developmental	124	Smart-Seq2	2016
Pollen	SRP041736	Human	cerebral cortex	Developmental	301	SMARTer	2014
Usoskin	GSE59739	Mouse	Brain	Healthy	622	STRT-Seq	2014
Macosko	GSE63473	Mouse	Retina	Healthy	44808	Drop-Seq	2015
Zheng-68 k	SRP073767	Human	PBMCs	Healthy	68579	10x Genomics	2017

Zheng-73 k	SRP073767	Human	Peripheral Blood	Healthy	73233	10x Genomics	2017
Darmanis	GSE67835	Human	Brain	Healthy	466	Smart-Seq2	2015
Pancreatic	GSE73727	Human	Islet Cells	Healthy	60	Smart-Seq2	2016
Silver	GSE115189	Human	PBMCs	Healthy	2590	10x Genomics Chromium	2018
Zhengmix4eq	SRP073767	Human	Peripheral Blood	Healthy	3994	10x Genomics	2016
ACINAR	ACINAR	Human	Prostate tissue	Disease	3043	10x Visium	2022
ATAC	GSE129785	Human	PBMCs	Healthy、Developmental	7034	10x scATAC-seq	2019
BATCH	GSE132044	Human	PBMCs	Healthy	6276	10x scRNA-seq （V2/V3）	2019
CITE	GSE100866	Human	CBMCs	Healthy	8617	CITE-seq	2017
EMBYRO	E-MTAB-3929	Human	Embryonic cells	Developmental、Healthy	1289	scRNA-seq	2016
IFNB	GSE96583	Human	PBMCs	Disease+Treatment	13999	10x scRNA-seq	2017
MARROW	GSE72857	Mouse	Bone marrow	Developmental	2660	MARS-seq	2015
MOBSC	GSE121891	Mouse	olfactory bulb	Health+Sensory manipulation	12640	10x Chromium	2018
OVARIAN	OVARIAN	Human	ovarian	Disease	3455	10x Visium	2022
PANCREAS	GSE132188	Mouse	pancreatic	Developmental	2087	10x scRNA-seq	2019

SCGEMMETH	SRP077853	Human	foreskin fibroblasts	Developmental	142	scGEM	2016
SCGEMRNA	SRP077853	Human	foreskin fibroblasts	Developmental	177	scGEM	2016
SCIATAC	GSE111586	Mouse	bone marrow	Healthy	4025	sci-ATAC-seq	2018
SLIDE	SCP354	Mouse	coronal cerebellum	Healthy	23372	Slide-seq	2021
VISIUM	SeuratData	Mouse	Brain	Healthy	2696	10x Visium	2019
Nakamura	GSE74767	macaque	Embryonic cells	Developmental	182、82、272、348	SC3-seq	2016
Horns	GSE100058	fly	Olfactory projection neurons	Developmental	277、168、454	smart-seq2	2017
Plass	GSE103633	planaria	whole organism level	Healthy	4288、4341、2270、4003、2338、2349、10578、10653、8589、8960、1986、694、292	drop-seq	2018

Table S3: Synthetic Datasets parameters.

Simulation Dataset	Cell Type	Cell	Gene	Dropout.mid	Batch.number	Batch.facLoc Batch.facScale	De.prob	De.facLoc De.facScale	Probability of Outliers
1	5	2000	10000	/	/	/	0.1	0.1,0.4	0.05
2	7	2000	10000	/	/	/	0.1	0.1,0.4	0.05
3	9	2000	10000	/	/	/	0.1	0.1,0.4	0.05
4	11	2000	10000	/	/	/	0.1	0.1,0.4	0.05
5	13	2000	10000	/	/	/	0.1	0.1,0.4	0.05
6	15	2000	10000	/	/	/	0.1	0.1,0.4	0.05
7	5	100	10000	/	/	/	0.1	0.1,0.4	0.05
8	5	500	10000	/	/	/	0.1	0.1,0.4	0.05
9	5	1000	10000	/	/	/	0.1	0.1,0.4	0.05
10	5	5000	10000	/	/	/	0.1	0.1,0.4	0.05
11	5	10000	10000	/	/	/	0.1	0.1,0.4	0.05
12	5	20000	10000	/	/	/	0.1	0.1,0.4	0.05
13	5	30000	10000	/	/	/	0.1	0.1,0.4	0.05
14	5	40000	10000	/	/	/	0.1	0.1,0.4	0.05
15	5	50000	10000	/	/	/	0.1	0.1,0.4	0.05
16	5	2000	5000	/	/	/	0.1	0.1,0.4	0.05
17	5	2000	20000	/	/	/	0.1	0.1,0.4	0.05

18	5	2000	30000	/	/	/	0.1	0.1,0.4	0.05
19	5	2000	40000	/	/	/	0.1	0.1,0.4	0.05
20	5	2000	50000	/	/	/	0.1	0.1,0.4	0.05
21	5	2000	10000	-1	/	/	0.1	0.1,0.4	0.05
22	5	2000	10000	0	/	/	0.1	0.1,0.4	0.05
23	5	2000	10000	1	/	/	0.1	0.1,0.4	0.05
24	5	2000	10000	2	/	/	0.1	0.1,0.4	0.05
25	5	2000	10000	3	/	/	0.1	0.1,0.4	0.05
26	5	2000	10000	/	2	/	0.1	0.1,0.4	0.05
27	5	2000	10000	/	4	/	0.1	0.1,0.4	0.05
28	5	2000	10000	/	6	/	0.1	0.1,0.4	0.05
29	5	2000	10000	/	8	/	0.1	0.1,0.4	0.05
30	5	2000	10000	/	10	/	0.1	0.1,0.4	0.05
31	5	2000	10000	/	2	0.2	0.1	0.1,0.4	0.05
32	5	2000	10000	/	2	0.4	0.1	0.1,0.4	0.05
33	5	2000	10000	/	2	0.6	0.1	0.1,0.4	0.05
34	5	2000	10000	/	2	0.8	0.1	0.1,0.4	0.05
35	5	2000	10000	/	2	1.0	0.1	0.1,0.4	0.05
36	5	2000	10000	/	/	/	0.05	0.1,0.4	0.05
37	5	2000	10000	/	/	/	0.15	0.1,0.4	0.05
38	5	2000	10000	/	/	/	0.2	0.1,0.4	0.05
39	5	2000	10000	/	/	/	0.25	0.1,0.4	0.05
40	5	2000	10000	/	/	/	0.3	0.1,0.4	0.05
41	5	2000	10000	/	/	/	0.1	0.2	0.05
42	5	2000	10000	/	/	/	0.1	0.4	0.05

43	5	2000	10000	/	/	/	0.1	0.6	0.05
44	5	2000	10000	/	/	/	0.1	0.8	0.05
45	5	2000	10000	/	/	/	0.1	1.0	0.05
46	5	2000	10000	/	/	/	0.1	0.1,0.4	0.1
47	5	2000	10000	/	/	/	0.1	0.1,0.4	0.2
48	5	2000	10000	/	/	/	0.1	0.1,0.4	0.3
49	5	2000	10000	/	/	/	0.1	0.1,0.4	0.4
50	5	2000	10000	/	/	/	0.1	0.1,0.4	0.5

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